Reg. No. | | | | | | | | | | |

[14ME82]

# EIGHTH SEMESTER B.TECH EXAMINATION, JUNE - 2019

## POWER PLANT ENGINEERING

**Time: 3 Hours** **Maximum Marks:70**

**Instructions:**

1. Answer any **FIVE** full Questions choosing at least TWO from each part.
2. Missing data may be suitably assumed.
3. Draw neat sketch wherever necessary.

### PART - A

1. a. Discuss the resources and development of power in India. **08**

   b. List out any six characteristics of ash handling plant. **06**

2. a. What are the different methods of firing coal? What are the advantages of mechanical methods of firing coal? Discuss the principle of underfeed stokers. **10**

   b. What is a natural draught cooling tower? What is the reason of its hyperbolic shape? **04**

3. a. Give a brief classification of cooling in diesel engines. Discuss the principle of evaporative cooling. **08**

   b. Write short notes on super charging. **06**

4. a. With P-V and T-S diagram explain the working of open cycle gas turbine plant. **06**

   b. With neat sketch explain combined gas cycle power plant. **08**

### PART – B

5. a. Write short notes on **06**

   i. Hydrograph
   ii. Flow duration curve
   iii. Mass curve

   b. What do you understand by run-off river plant? How its performance is increased by introducing a pondage in the plant? **08**

6 a. With neat sketch explain working principle of horizontal axis wind mill. **08**

   b. Discuss the principle of solar pond. **06**

7. a. What is a nuclear reactor? Describe the various parts of a nuclear reactor. **06**

   b. Discuss different methods of radioactive waste disposal **08**

8. a. Define the following; **08**

   i. Load curves
   ii. Demand factor
   iii. Connected load
   iv. Diversity factor

   b. Discuss the methods employed to control pollution **06**

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